LESSON PLAN

LOW SPEED FLIGHT (9)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	03
Total No. of Lesson Plans	13	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Alighting gear, basic design requirements, types of u/c and functions
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Alighting gear, basic design requirements, types of u/c and functions.
5. Specific Objectives	9.1.1.1 Trainee will be able to describe various Aircraft engineering systems.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of Landing Gear System Diagrams on Wall Charts; Instructor supervision during Q&A; session.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the landing gear?
(b) Name two basic design requirements for landing gear.
(c) List the three main types of landing gear configurations.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Basic design requirements of landing gear • Shock absorption mechanisms • Braking systems and wheel types • Steering and nose gear functions • Weight distribution and load limits	L & Ds	PPT, IB	15 min
(b)	• Tricycle landing gear configuration • Tail-dragger landing gear configuration • Bicycle landing gear configuration • Functions of main and nose gear • Retraction and extension mechanisms	L & Ds	PPT, PDF	20 min
(c)	• Hydraulic vs mechanical retraction systems • Tire pressure and inflation standards • Wheel brakes and anti-skid systems • Gear door operation and sealing • Maintenance access points	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between tricycle and tail-dragger configurations.
(b) Describe the role of shock absorbers in landing gear design.
(c) Identify the function of the nose gear in a tricycle configuration.

12. Home Assignment and Follow Up	Prepare a brief note on the typical circuit for retraction and extension of landing gear to be discussed in LP 04.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Safety and emergency devices for retraction and extension, typical u/c circuits, brief servicing aspects.	Refer to the master syllabus reference list	LP 04

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min